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About Protocol Studio

About Protocol Studio
Protocol Studio is a display tool that shows high-level protocol-based exchanges with the DUT in protocol transaction format rather than in 1's and 0's format. This lets you quickly verify that a protocol is implemented correctly and that the correct transactions are being exchanged with the DUT.
Protocol Studio lets you debug protocol-based transactions with the DUT in "mission mode," that is, communicating with the DUT using the protocols that are designed into the DUT. Protocol Studio supports nWire, DDR, and PCIe transactions.
The following topics are available:

•	Using Protocol Studio

•	Port Debug Display

•	Protocol Studio Toolbar and Command Reference

IG-XL includes additional Protocol Aware tools:

•	Protocol Definition Editor:

  - Define a protocol from scratch
  - Edit an existing protocol definition
  - Graphically view frame waveforms
- CyclePro:
  - Raw cycle display mode
  - Symbol10b display mode
  - PCIe Link traffic decode (CMEM) and ATE Tx traffic decode (Tx HRAM)
  - Layer Selection: Transaction, Data Link, PHY

For more information, see:

- DDR PA Debugging
- nWire PA Debugging
- PCIe PA Debugging
- sWire PA Debugging
- About Protocol Definition Editor
- About CyclePro

For information about Protocol Aware in general and nWire in particular, see the following:

- About Protocol Aware

- nWire Protocol Aware

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ProtocolStudioPortDebugDisplay

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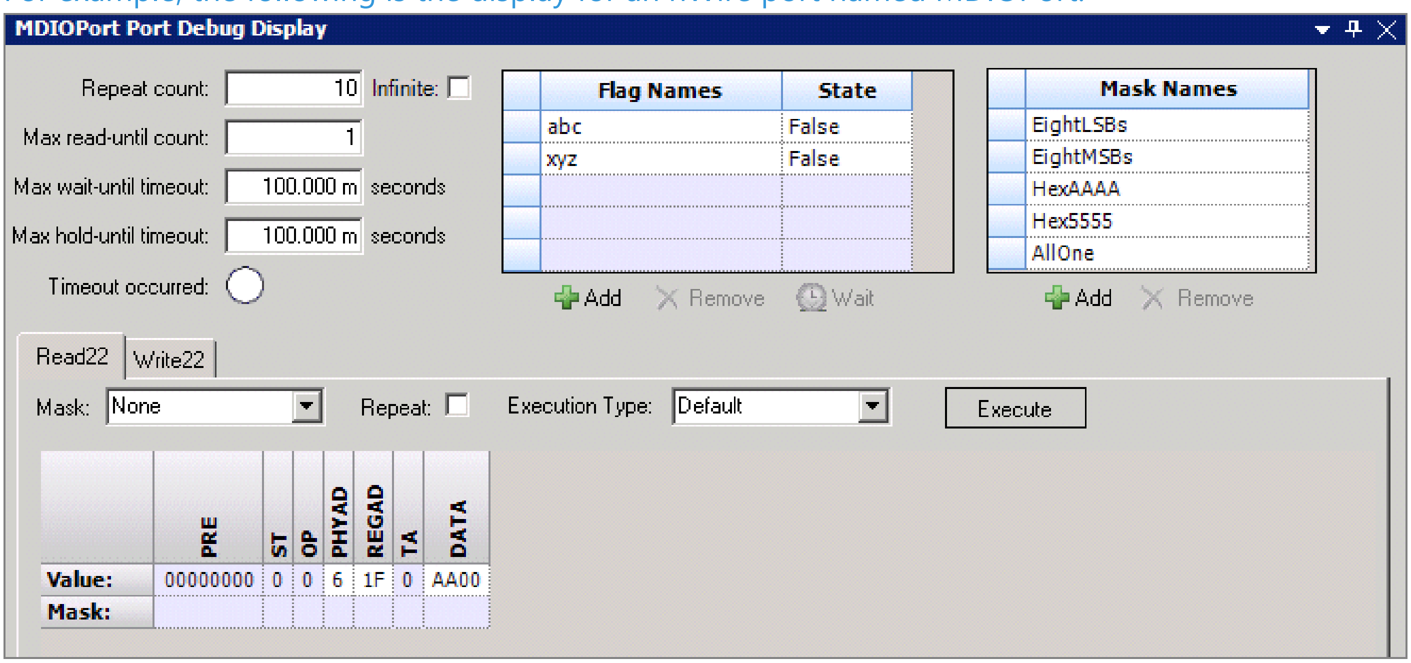
Port Debug Display

Port Debug Display

The Port Debug Display panel shows the debug display for the specified port. You can modify the transaction parameters and re-execute it.

To bring up a specific Port Debug Display, first select the port in the Ports/Modules Explorer or select a transaction in the main grid. Then right-click and select Show Port Debug Display from the context menu.

For example, the following is the display for an nWire port named MDIOPort:



Port Debug Display Showing an nWire Port

You can perform certain actions using the debug display. You can specify whether the action should apply to the currently selected site or to all sites.

Use the All button, which is next to the Master button. When enabled, it specifies that any action performed by the debug display is to be applied to all sites, not just the currently selected site.

The Port/Module actions affected by this control include:

- Enable port
- Disable port
- Enable port timeout
- Set port timeout value
- Halt port
- HaltWait port
- IdleWait port
- Set module as default

The following Port/Module query actions are always performed only against the displayed site:

- Query port status
- Query enabled state
- Query timeout value, timeout enabled

The following Port/Module actions are not site-aware:

- Query port family
- Query port type
- Load module file

- Unload module
- Reload module file
- Unload all module files
- Set module execution mode

For more information on the use of the display with specific protocols, see:

- DDR PA Debugging
- nWire PA Debugging
- PCIe PA Debugging
- sWire PA Debugging

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Protocol Studio Toolbar and Command Reference

Protocol Studio Toolbar and Command Reference

This section lists actions that are available from the Protocol Studio menus and toolbars. References to more complete descriptions elsewhere in Help are included.

Most toolbar buttons correspond to menu commands, but some are available only on the toolbars.

The section is organized on where the command appears, on the toolbar or on one of the menus:

- Toolbars
- File Menu Commands
- View Menu Commands
- Tools Menu Commands

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Protocol Studio Toolbar and Command Reference : Toolbars

Toolbars

The two Protocol Studio toolbars are made up of groups of tools.

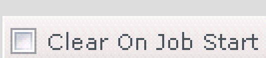
The first toolbar looks like this:

First Protocol Studio Toolbar

The first few buttons apply to all data in the display:

	Selects the site, either using the master site as set in DataTool or selecting a site. All specifies whether Port Debug Display actions apply to all sites or the currently selected site. See Selecting the Site.
	View>Refresh or F5 or F12 Refreshes the display. This is needed when Protocol Studio's title bar indicates that a refresh is required. See Refreshing the Display.

The rest of the buttons on the first toolbar apply to the Main Grid:

	Whether to have Protocol Studio automatically update the display when new captures occur. See Refreshing the Display.
	View>Clear Clears all transactions from the display. See Clearing Transactions.
	Whether to clear the display whenever the job starts. See Clearing Transactions.

Data Words: 2	How many data words to display in the Main Grid before using the Data window. See Data Window.

The second toolbar contains buttons for controlling specific aspects of the Protocol Studio display:

Second Protocol Studio Toolbar

The first group of buttons control the appears of columns. The corresponding commands all appear on the View menu:

Hide	Hides the selected columns. See Hide and Unhide.
Filter	Shows only those rows with a specified value in the selected column. See Filter.
Color	Uses the selected background color for rows with a specified value in the selected column. See Color.

The second group of buttons controls how data appears in the columns of the Main Grid and the Details and Data windows:

Radix:	Selects the format for displaying the data: Hex, Octal, Binary, Decimal, ASCII.
Literals	Whether to display the literals the protocols define or their numeric value. See Data Display Options.
Data Columns: 16	How many columns to use to display the data in the Data window. See Data Window.

The final item lets you navigate to the first, previous, next, or last failure:

Protocol Studio Navigation Controls

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Protocol Studio Toolbar and Command Reference : File Menu Commands

File Menu Commands

File commands let you debug offline and exit from Protocol Studio.

Open	Opens a previously saved transaction file for offline debugging. See Saving Transactions to a File.
Save	Saves the currently captured transactions to a file that was most recently saved to. If this is the first save, it acts as Save As. See Saving Transactions to a File.
Save As	Saves the currently captured transactions to the specified folder and file. If the new file already exists, you are asked if you want to overwrite the file. See Saving Transactions to a File.
Exit	Exits from Protocol Studio.

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Protocol Studio Toolbar and Command Reference : View Menu Commands

View Menu Commands

Messages	Shows the Message Display Panel. See About the Message Panel.
View and Port Setup	Shows the View and Port Setup panel. See View and Port Setup.
Details	Shows the Details window. See Main Grid and Details Window.
Port Debug Display	Shows the Port Debug Display.
Data	Shows the Data window. See Data Window.
Properties	Shows the Properties panel. See Properties Panel.
Capture Setup	Shows the Capture Setup panel.
Vertical Headers	Toggles the column headers between horizontal (default) and vertical. This affects all headers in the Main Grid and in the Details panel.
Gray Out Empty Fields	Grays out fields with no data for ease of reading.
Hide Columns	 Removes selected columns from the Main Grid. A pull-right submenu lists currently displayed columns. See Hide and Unhide.
Unhide Columns	Redisplays selected column in the Main Grid. A pull-right submenu lists currently hidden columns. See Hide and Unhide.
Take Snapshot	Takes a snapshot of the current state of the tool. See Taking and Viewing Snapshots.
View Snapshot	Displays a previously taken snapshot.

Filter	Shows only those transaction rows that have the selected value for the selected column. See Filter.
Color	Highlights with the specified background color the transaction rows that have the selected value for the selected column. See Color.
Clear	Clears all data from the display. See Clearing Transactions.
Refresh	F5 or F12 Refreshes the display. Protocol Studio indicates when a manual refresh is needed. See Refreshing the Display.
Restore Design View	Restores the Protocol Studio window to its initial default state.

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Protocol Studio Toolbar and Command Reference : Tools Menu Commands

Tools Menu Commands

Waveform Display	Brings up the Waveform Display tool.
Connect to IG-XL	Reconnects to the IG-XL workbook from stand-alone mode or after disconnecting by loading a saved file. See Saving Transactions to a File.

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Using Protocol Studio

Using Protocol Studio

Protocol Studio is an integrated tool that can display executed transactions, including transactions issued interactively. Interactive transactions are those executed by a VBT transaction script, executed through a Port Debug Display, or issued at the immediate window.

The following topics are available:

- Display Basics
- Data Window
- Column Display Options
- Properties Panel
- Capture Setup
- View and Port Setup
- Saving Transactions to a File

Note: Connected and Disconnected Modes

In most instances, you use Protocol Studio with a running test program and IG-XL workbook. You run to a breakpoint and step through the program, using Protocol Studio to view the resulting transactions.

Protocol Studio also lets you save captured transactions to a file and then later open the file to examine the transactions without the tester. When you open such a file, you are disconnected from IG-XL, and Protocol Studio runs in essentially stand-alone mode. You can reconnect to IG-XL at any point. See Saving Transactions to a File.

The descriptions in this section assume that Protocol Studio is connected to IG-XL. For more information about using Protocol Studio when it is not connected to IG-XL, see Saving Transactions to a File.

Note: Complete Pattern and Module Execution Before Running Protocol Studio

When executing PA transaction modules from digital patterns, you must program your VBT code so that both the digital pattern and the PA module have finished executing before processing results. This ensures that the Protocol Studio lists transactions in the correct order when stepping through the test program. Use the following VBT syntax in this sequence to ensure that pattern and module execution is complete:

1.	TheHdw.Digital.Patgen.HaltWait: Ensure that the digital pattern has finished executing by waiting for the running PatGen to halt.

2.	TheHdw.Protocol.Ports.IdleWait: Ensure that any PA modules started by the digital pattern have finished executing by waiting for the PA engines to be in an idle or stopped state.

☑ **Note:**

HaltWait must come before IdleWait in the VBT code for Protocol Studio to work correctly.

For more information, see Executing PA Transaction Modules From Digital Patterns.

Note: CMEM Capture Mode

When you are using Protocol Studio with the PCIe protocol, set CMEM capture mode to Payload. If CMEM capture mode is set to Cycles, no data is displayed in the Protocol Studio window.

See PCIe PA CMEM Programming.

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Using Protocol Studio : Display Basics

Display Basics

The section covers the basic of viewing and controlling the Protocol Studio display:

- Opening Protocol Studio
- Main Grid and Details Window
- Failing Transactions
- Data Display Options
- Navigation
- Refreshing the Display
- Clearing Transactions
- Message Panel
- Taking and Viewing Snapshots
- Selecting the Site

- Grid Font Zoom

See also the following topics that are covered in other sections:

- Data Window
- Column Display Options
- Properties Panel
- Capture Setup
- View and Port Setup
- Saving Transactions to a File

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Using Protocol Studio : Display Basics : Opening Protocol Studio

Opening Protocol Studio

You can open Protocol Studio by clicking Protocol Studio in the One-Click Launch group on the IG-XL Tools tab.

You can also open Protocol Studio from PatternTool, Serial Protocol Display, and Waveform Display.

To open Protocol Tool for stand-alone use, in the Start menu select the following:

Start>All Programs>Teradyne IG-XL Vversion_uflx>Tools>Protocol Definition Editor

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Using Protocol Studio : Display Basics : Main Grid and Details Window

Main Grid and Details Window

Protocol Studio presents an ordered sequence of captured transactions, one transaction per line. The complete display of transactions uses both the Main Grid and a Details window:

Index	Port	Frame	Timestamp	Repeat Index	Execution Order	fn	Payload	offset	bar	size
- 1 >	pcie_ep	MWr	0.000e00	0	0	Fn0	FFFFFFFF	8	Bar0	
2 <	pcie_ep	Cmpl	0.000e00	0	0					
- 3 >	pcie_ep	MRd	0.000e00	0	0	Fn0	FFFFFFFF	8	Bar0	4
4 <	pcie_ep	CmplD	0.000e00	0	0		FFFFFFFF			
+ 5 >	pcie_ep	MWr	0.000e00	0	0	Fn0	FFFFFFFF	8	Bar0	
+ 7 >	pcie_ep	MRd	0.000e00	0	0	Fn0	FFFFFFFF	8	Bar0	4

Details

Index	Port	Frame	Timestamp	Repeat Index	Execution Order	fn	poison	fbEnables	Payload	offset	ba
7	pcie_ep	MRd	0.000e00	0	0	Fn0	False	Enable1111	FFFFFFFF	8	Bar

Protocol Studio, Main Grid and Details Window

In the main display, Protocol Studio shows the most important information about a transaction: the port and frame names; timestamp, repeat index, and execution order; and the transaction's fields of interest, those that you have programmed or that contain data from the DUT.

The Details window displays all information from the transaction. (If a cell is blank, the Details window does not display it, even if it appears in the Main Grid.) Click a transaction row in the Main Grid to bring up the Details window with all transaction information.

Note these points about the Main Grid:

- The Index column contains the index number followed by an arrow that points left, right, or up. This indicates the direction of the transaction, from the DUT's point of view:
  - > (right arrow): transaction is to the DUT

  - < (left arrow): transaction is from the DUT
  - ^ (up arrow): direction is unknown or cannot be determined

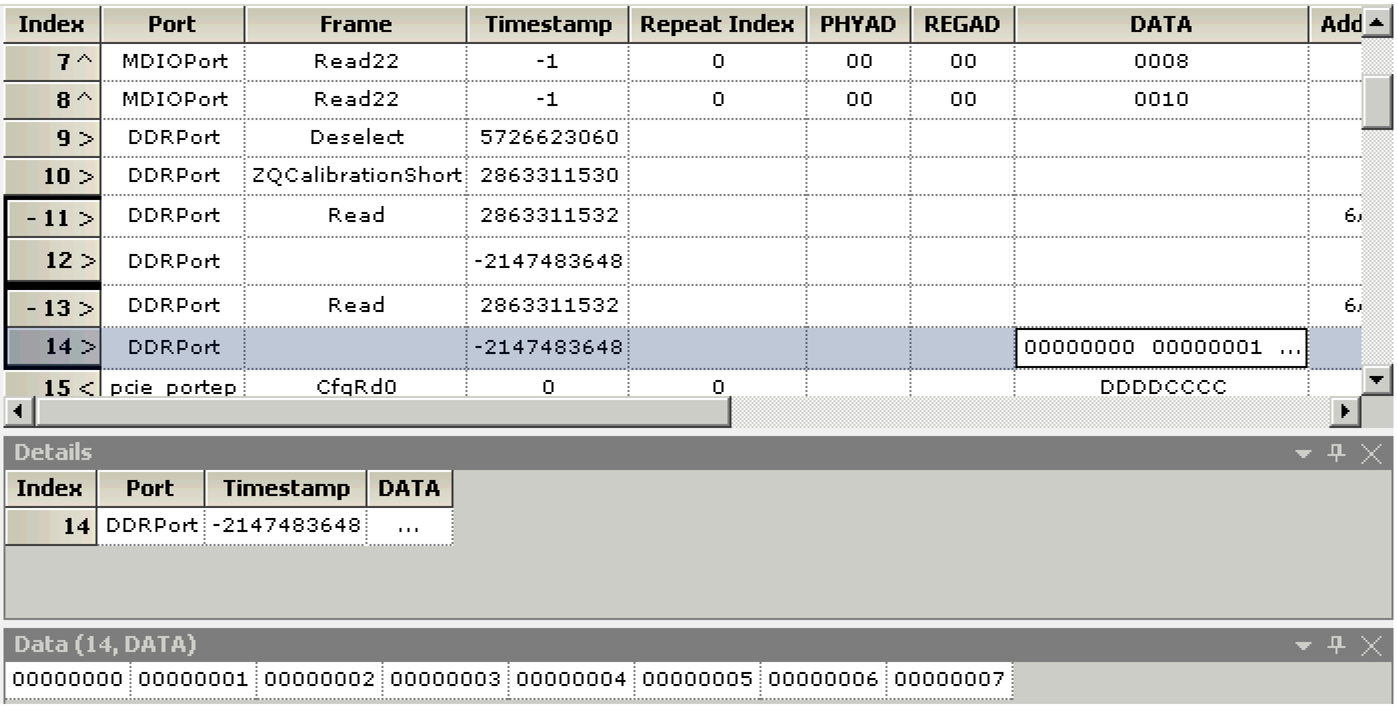
- If the protocol supports subtransactions, the Index column includes a plus sign (+) to indicate that the transaction is made up from multiple subtransactions. Click + to view all subtransactions.
- 

The display uses a waveform symbol to indicate a transaction on which a trigger occurred.
- If the column is not wide enough to display the data, an ellipsis (...) appears in the middle of the data:

fn	Payload	offset
Fn0	1111111...1111111	8
Fn0	1111111...1111111	8
	1111111...1111111	

Ellipsis in a Narrow Column
This usually happens only if you manually resize a column.

- If an ellipsis appears at the end of a data field, in either the Main Grid or the Details window, this means that you have specified that data fields of this length should be displayed in the Data window:



Ellipsis in a Data Field

For more information, see Data Window.

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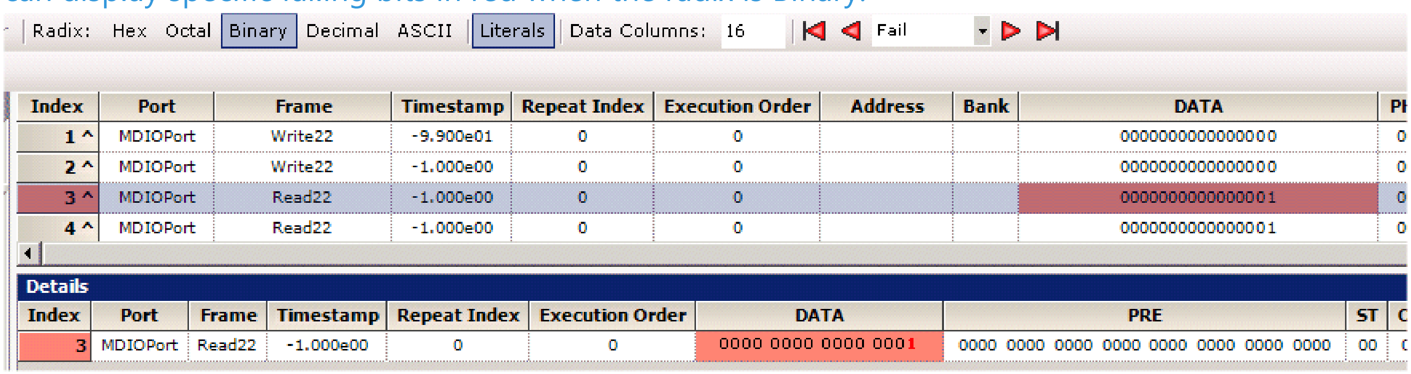
Using Protocol Studio : Display Basics : Failing Transactions

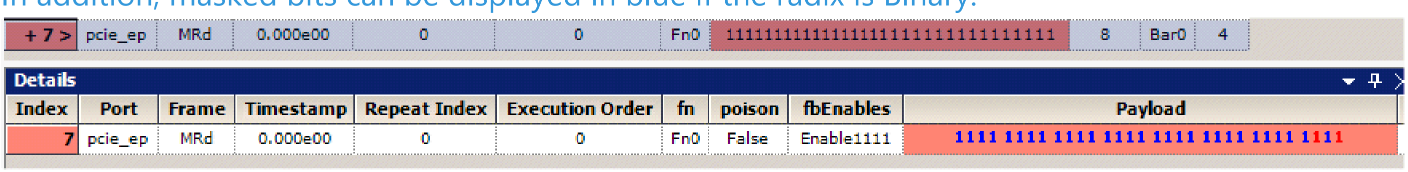
Failing Transactions

Protocol Studio highlights failing transactions by coloring the row's Index column red. It also highlights the field where the failure occurs. If the protocol supports failing bits, the Details window can display specific failing bits in red when the radix is Binary:



Failing Transactions Displayed in Red

In addition, masked bits can be displayed in blue if the radix is Binary:



Masked Bits Displayed in Blue

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Using Protocol Studio : Display Basics : Data Display Options

Data Display Options

You can control how the data is displayed in the Main Grid, the Details window, and the Data window, using these buttons:

Radix: Hex Octal Binary Decimal ASCII | Literals

Data Control Buttons

Radix	These buttons let you display data in hex, octal, binary, decimal, or ASCII format. Binary format is useful when displaying bit-level information in the Details window, such as failing bits.
Literals	This button specifies whether certain fields should be displayed as bits or as the literals that the protocol has defined for these fields. Click this button to use literals.

The following data is displayed using literals:

bar	tc	snoop	relaxedOrdering	lbEnables
Bar0	TC0	False	False	Enable0000

Fields Displayed as Literals

The following shows the same fields with literals turned off:

bar	tc	snoop	relaxedOrdering	lbEnables
0	0	0	0	00

Fields Displayed with Literals Turned Off

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Using Protocol Studio : Display Basics : Navigation

Navigation

The navigation controls let you go to the first previous, next, or last failure in the display:

Navigation Controls

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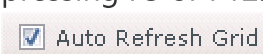
Using Protocol Studio : Display Basics : Refreshing the Display

Refreshing the Display

Protocol Studio must be refreshed when new transactions are captured or when certain events occurs that invalidate the current display: for example, executing a test that starts different ports from the previous test. You can either manually refresh the display or turn on autorefresh.

When a refresh is needed, Protocol Studio shows a message in the window title bar. You can manually refresh the display by clicking this button, using the command View>Refresh, or by pressing F5 or F12.



When issuing transactions interactively, by stepping through a job, you have the option of turning on autorefresh by checking this box.

If autorefresh is turned on, Protocol Studio automatically displays new transactions as they are executed.

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Using Protocol Studio : Display Basics : Clearing Transactions

Clearing Transactions

Protocol Studio accumulates transactions as they are executed. If you run a job multiple times, it can continue to accumulate them.

At any point, you can all currently displayed transactions from the display and from the tool memory. Use this button or the View>Clear command.

You can also check this box to have Protocol Studio automatically clear the display each time the test program is started.

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Using Protocol Studio : Display Basics : Message Panel

Message Panel

Messages that do not require an immediate user response are written to the Message Panel. For more information, see About the Message Panel.

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Using Protocol Studio : Display Basics : Taking and Viewing Snapshots

Taking and Viewing Snapshots

Protocol Studio lets you take snapshots of the current state of the tool. Later you can view the snapshots for a visual comparison of the tool states.

The command View>Take Snapshot takes a snapshot of the Protocol Studio window. The View>View Snapshot command lets you view the snapshots. It has these options:

Most Recent	Displays the most recent snapshot.
Other	Brings up a browser to let you select the snapshot to view.

The display uses the default Windows image viewer.

By default, a snapshot is saved to the test program folder. If the default folder is read-only, it is saved to the tmp folder. You are not prompted to specify a folder. The snapshot is saved as JPEG file with a filename of this form: ProtocolStudio__YYMMDDHHMMSS.jpg. This means that snapshots in the browser will be in chronological order.

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Using Protocol Studio : Display Basics : Selecting the Site

Selecting the Site

Protocol Studio shows the data for one site at a time. You can use the Site selector to specify a site:

Protocol Studio Site Selection

If you click Master, Protocol Studio uses the master site as it is defined in DataTool. The site number displayed the current master site, but is grayed out because you cannot change it.

If you disable Master, you can manually select the site to be displayed.

If you have enabled Master and the master site changes in IG-XL, Protocol Studio indicates that you must refresh the display.

The All button, next to the Master button, is used with the Port Debug Display. When enabled, it specifies that any action performed by the debug display is to be applied to all sites, not just the currently selected site.

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Using Protocol Studio : Display Basics : Grid Font Zoom

Grid Font Zoom

The bottom of the Protocol Studio window has a sliding zoom control:

Protocol Studio Zoom Control

The control increases or decreases the size of the font in the display. As you move the slider, a box appears above the control showing the font size specified by the current position.

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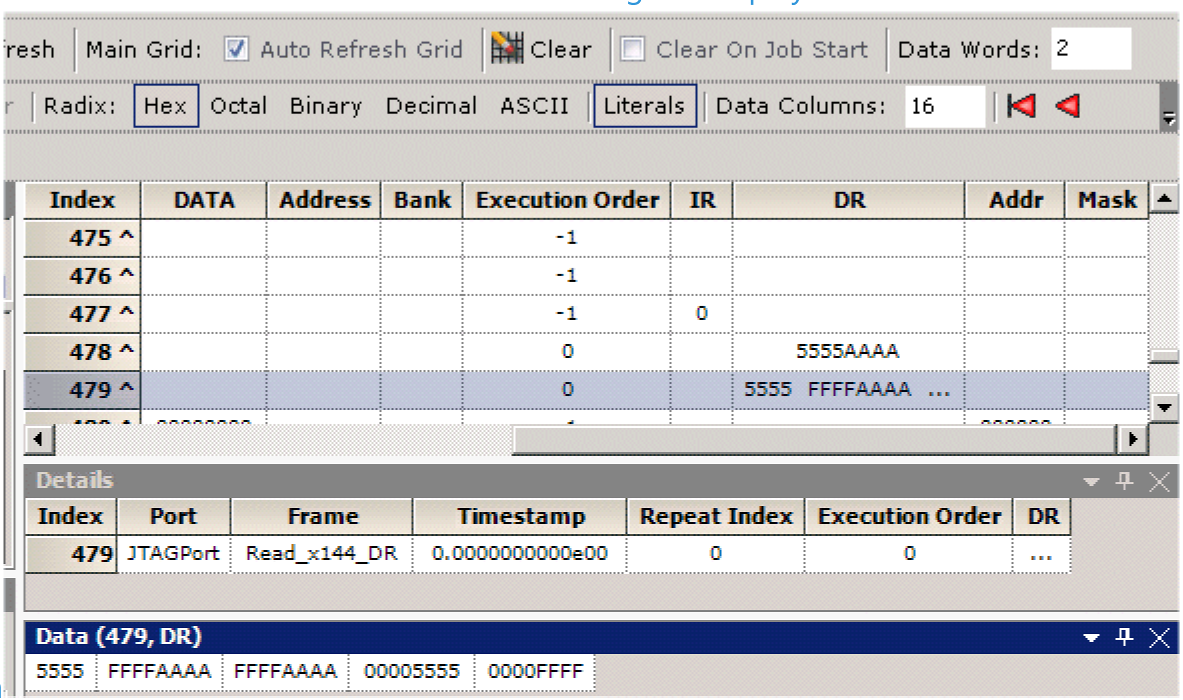
Using Protocol Studio : Data Window

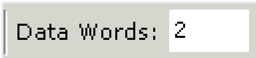
Data Window

Some very long data fields could make the columns in the Main Grid and Details window difficult to read, because you would need to scroll to see all the columns. To avoid this problem, Protocol Studio supports a Data window that displays long data as well as controls to let you specify how the Data window should be used. The following is a display that includes the Data window:

m 

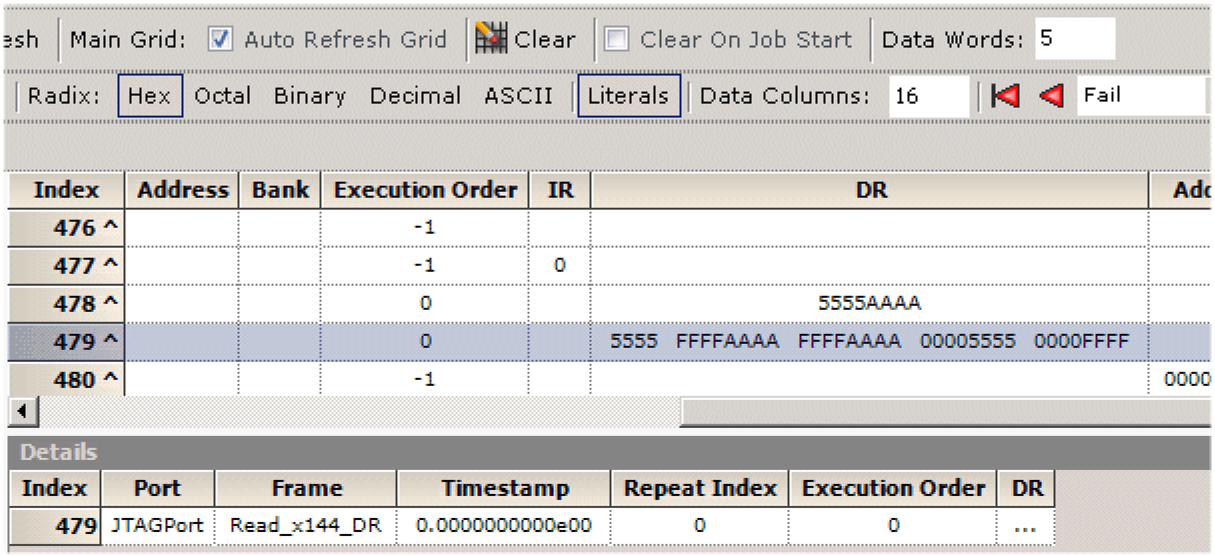
Data Window in the Protocol Display, Two Data Words Displayed

The DR cell in row 479, in either the Main Grid or the Details window, has an ellipsis at the end, indicating that there is more data to display. When you click the cell, the Data window comes up displaying all the data. The title of the Data window identifies the row number and column name of the related cell.



You control when data is displayed in the Data window and how it is displayed there. The Data Words control specifies how many data words are to appear in the column before the Data window is used. Here it is set to 2. This means that only the first two data words of the data are to be shown in the cell in the Main Grid. As shown above, the two data words are in the cell, followed by the ellipsis.

If you change Data Words to 5, all the data is displayed without the need for an ellipsis in the Main Grid:

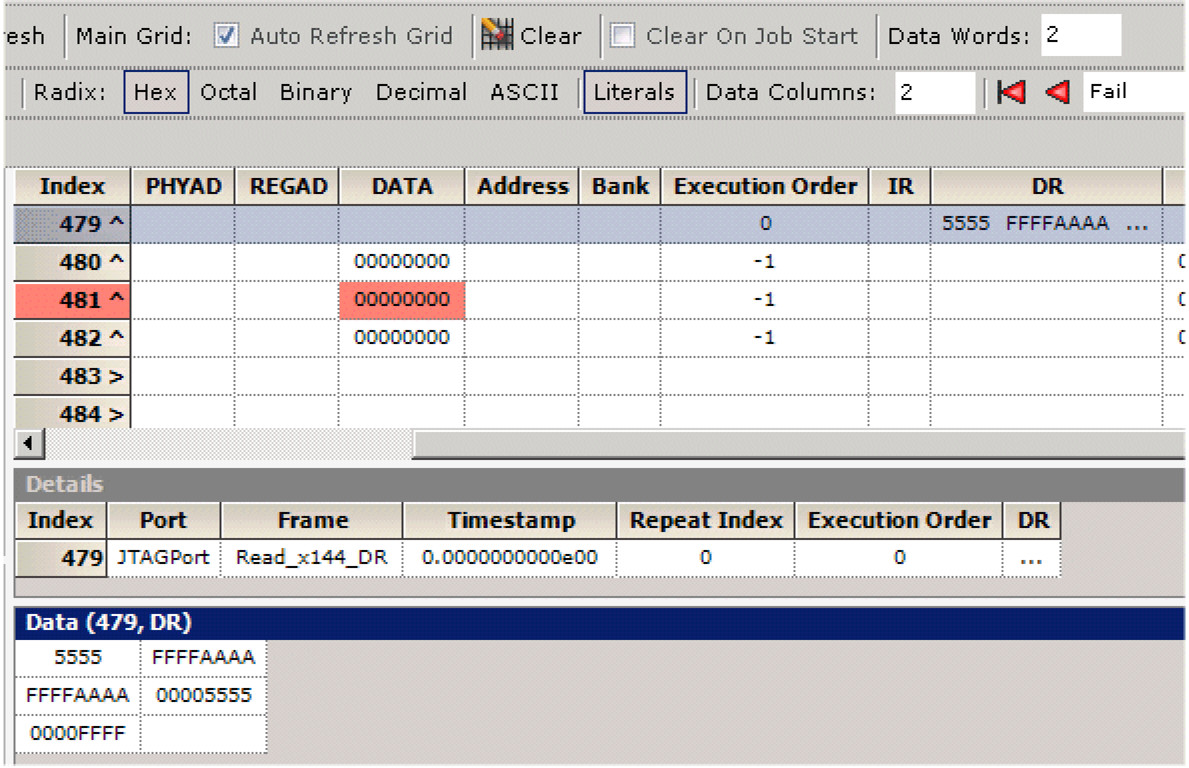


Data Window in the Protocol Display, Five Data Words Displayed

The Details window still shows the ellipsis. If you click in the cell, the Data Window comes up.

The complete data value is shown in the Data window. You can control how data is displayed there.

The Data Columns control specifies into how many columns the data should be divided. You can choose whatever number is the most convenient for reading. The first example above uses the default, 16. Because the Data Columns number is greater than the number of words in the data (5), five columns are shown. If you reduce the number to 2, the result is this:



Data Window in the Protocol Display, Two Columns Displayed

Note that an ellipsis in the middle of a cell indicates that the cell is too narrow and does not indicate that the Data window can be used. This usually happens because you have manually narrowed the column. To view the full data in this case, widen the column.

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Using Protocol Studio : Column Display Options

Column Display Options

Protocol Studio lets you control aspects of how the transactions are displayed. This section describes how to set nondefault display options:

- Vertical Header
- Filter
- Color
- Hide and Unhide

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Using Protocol Studio : Column Display Options : Vertical Header

Vertical Header

The width of column headers sometimes is greater than the width of the data in the columns. To get the minimum column width, you can switch the headers from the default horizontal to vertical display. Use the command View>Vertical Headers to toggle between vertical and horizontal headers.

The command affects all column headers, both in the Main Grid and in the Details panel.

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Using Protocol Studio : Column Display Options : Filter

Filter

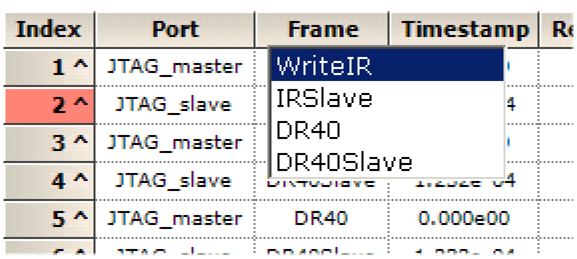
Protocol Studio lets you filter transactions to show only the ones of interest, that is, only those transaction rows that contain a particular value in one of the columns.

1. Select the column whose value you want to use as the filter by clicking the column header.

2. 

Invoke the Filter function. Use this button, the View>Filter command, or the context menu that comes up when you right-click the column header.

3. This brings up a list of all values in the column. Select the value you want displayed:



Filter Values Menu for a Column

The display now shows only those transaction rows that have the selected value in that column.

The filter also applies to future captures. Only those that match the filter will be shown.

To remove the filter, right-click any column header. From the context menu, select Remove current filter.

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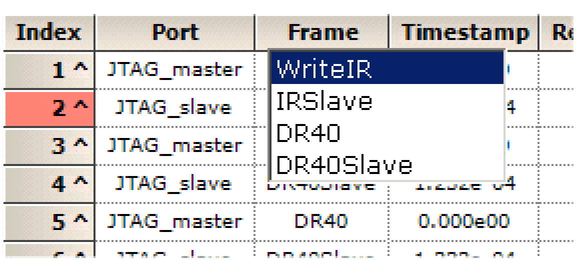
Using Protocol Studio : Column Display Options : Color

Color

You can use color to indicate all rows that have a particular value in a column. For example, you might want every row with a WriteIR frame to have the background color yellow. To do this, follow these instructions:

1. Select the column whose value you want to highlight by clicking the column header. In this example, this is the Frame column.
2.  Invoke the Color function. Use this button, the View>Color command, or the context menu that comes up when you right-click the column header.
3. This brings up a list of all the values in the column. Select the value you want colored:



Color Values Menu for a Column

4. A color chooser now comes up. Pick the color you want and click OK.

The result looks like this:

Index	Port	Frame	Timestamp	Repeat Index	Execution Order	IR	TDI_lower_32	readback_lower32	TDI_byte_1	readback_byt
1 ^	JTAG_master	WriteIR	0.000e00	0	0	000000				
2 ^	JTAG_slave	IRSlave	1.232e-04	0	0	000000				
3 ^	JTAG_master	DR40	0.000e00	0	0		00000000	00000000	00	00
4 ^	JTAG_slave	DR40Slave	1.232e-04	0	0					
5 ^	JTAG_master	DR40	0.000e00	0	0		00000000	00000000	00	00
6 ^	JTAG_slave	DR40Slave	1.232e-04	0	0					
7 ^	JTAG_master	WriteIR	0.000e00	0	0	000000				
8 ^	JTAG_master	DR40	5.760e-07	0	1		00000000	00000000	00	00
9 ^	JTAG_master	DR40	1.472e-06	0	2		00000000	00000000	00	00
10 ^	JTAG_master	DR40	2.368e-06	0	3		00000000	00000000	00	00
11 ^	JTAG_master	DR40	3.264e-06	0	4		00000000	00000000	00	00
12 ^	JTAG_master	DR40	4.160e-06	0	5		00000000	00000000	00	00
13 ^	JTAG_master	DR40	5.056e-06	0	6		00000000	00000000	00	00
14 ^	JTAG_master	DR40	0.000e00	0	0		00000000	00000000	00	00
15 ^	JTAG_slave	DR40Slave	1.232e-04	0	0					
16 ^	JTAG_master	WriteIR	0.000e00	0	0	000000				
17 ^	JTAG_master	DR40	5.760e-07	0	1		00000000	00000000	00	00
18 ^	JTAG_master	DR40	1.472e-06	0	2		00000000	00000000	00	00

Color Applied to Rows

The color also applies to future captures and will be applied to any new transaction row that match the color rule.

To remove a color, select the row header (the Index column) of a row that has the color. The right-click context menu has the option Remove row color rule. This removes the color from all the rows in the display.

You can apply more than one color rule. If you use different columns as the basis for the rules, colors may overlap. In this case, the most recently applied color is laid over the earlier colors. When removing color rules, if you select a row that has both colors applied, the more recently applied rule is removed first.

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Using Protocol Studio : Column Display Options : Hide and Unhide

Hide and Unhide

By default, Protocol Studio’s Main Grid shows a subset of all columns available for the transaction data, while the Details panel shows all columns. You can have the Main Grid show columns other than those shown by default. You can include columns that are by default not included or excluded columns that are included.

To hide a single column in the Main Grid, select it by clicking the column header. You can select more than one column at a time. Then either press this button or right-click the header and select Hide from the context menu. The selected columns are removed from the display.

The View>Hide Columns command has a pull-right submenu that lists all currently displayed columns. Select a column from the list to hide it.

To unhide columns, use the View>Unhide Columns command. Its pull-right submenu lists all currently hidden columns: both those that you have explicitly hidden and those that by default are not included in the Main Grid. You can unhide either kind of column.

The command View>Restore Design View returns columns to their default hidden and unhidden states.

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Using Protocol Studio : Properties Panel

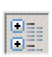
Properties Panel

The Properties panel displays the properties of the port, module file, or module that is currently selected in the Ports/Modules window. See Port/Modules.

If the Properties window is not visible, use View>Properties to display it.

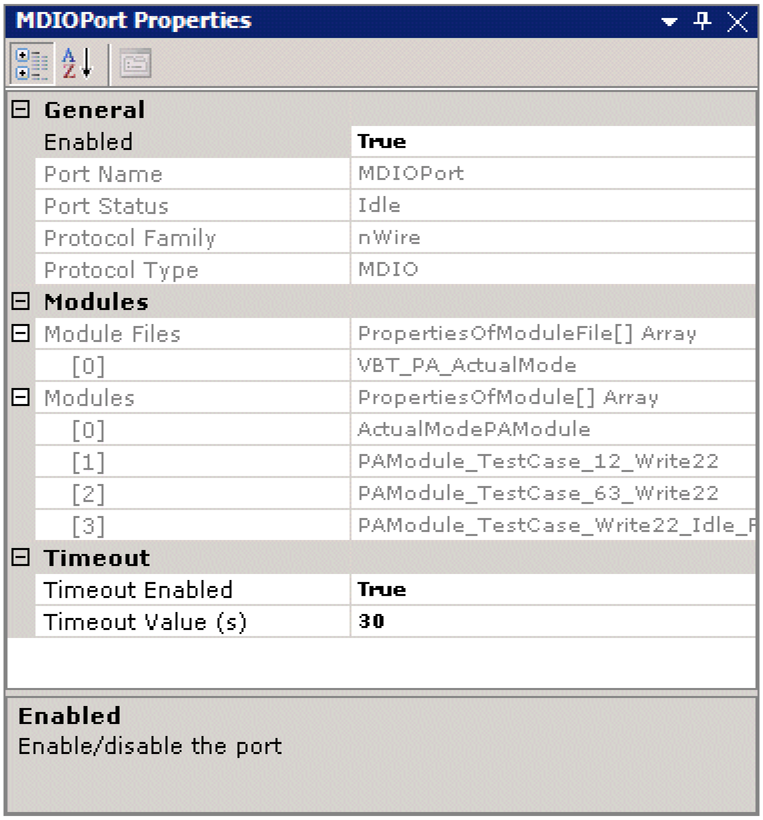
The Properties window can be divided into categories. The first category is always General. The other categories depend on the kind of item selected.

You can specify how properties are to be displayed using the buttons at the top of the Properties window:

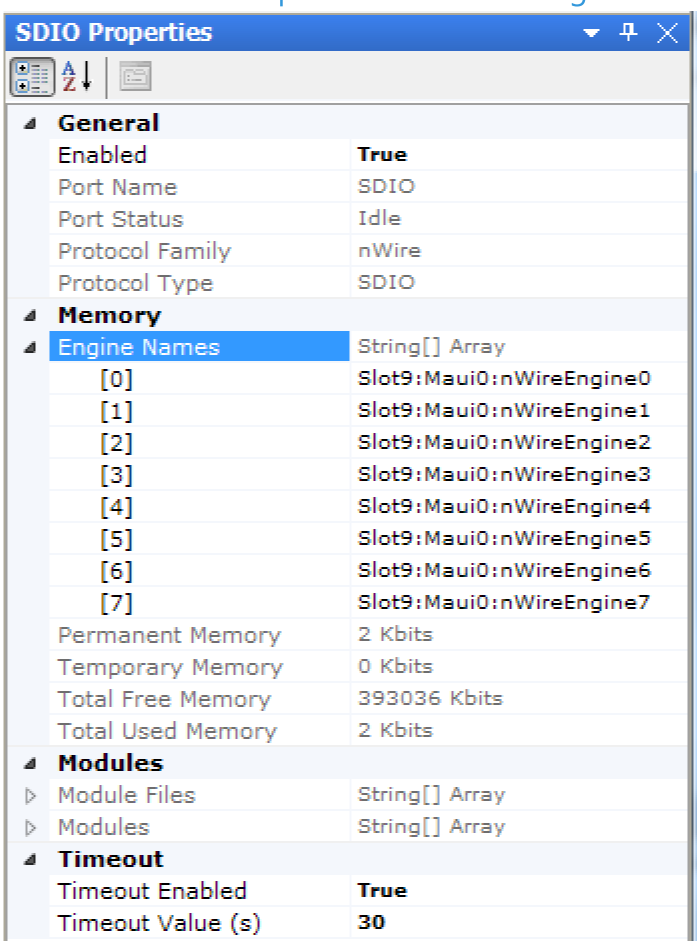
-  Categorized - for example, General, Modules, Timeout
-  Alphabetical, without category dividers

Properties are mostly read-only, although a few are read/write.

The properties listed depend on the item chosen in the Ports/Module window. The following shows the properties for a port:



Protocol Studio Properties Panel Showing Port Properties



Protocol Studio Properties Panel Showing Port Memory Properties (nWire Engines Only)

For a port, the properties displayed are:

General	Port name, status, and enabled state, and protocol family and type.
Memory	This group of properties is only supported for the nWire protocol family: • Engine Names: A list of nWire engines on the selected port. • Permanent Memory: Total allocated memory size of all permanent modules loaded in the selected port in kilobits. • Temporary Memory: Total allocated memory size of all temporary modules loaded in the selected port in kilobits. • Total Free Memory: Total available (free) engine memory for the selected port in kilobits. • Total Used Memory: Total memory size used for all loaded modules on the selected port in kilobits.
Modules	Any module files loaded for the port and the modules in the files.
Timeout	Timeout enabled state and value.

You can modify the port enabled state and the timeout enabled state and value.
For a module file, the properties displayed are: the module file's execution mode, file name, and port name; and the modules in the file. You can modify the execution mode.
For a module, the properties displayed are: the module's file name, name, and port, and whether it is the default module. You can modify the default state.

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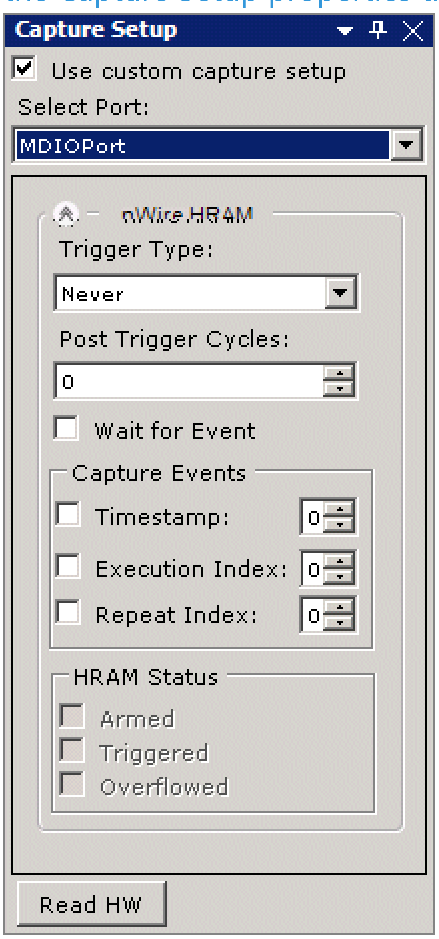
Using Protocol Studio : Capture Setup

Capture Setup

Part of the interactive debug process includes setting up HRAM triggering to capture transactions of interest. You do this using the Capture Setup panel. The panel displays the Capture trigger setup properties for the port selected. If the panel is not displayed, use View>Capture Setup to display it.

Capture properties are set to capture the maximum number of transactions by default.

Properties displayed in the Capture Setup panel depend on the instrument. For example, this shows the Capture Setup properties the nWire protocol:



Protocol Studio Capture Setup Panel Showing nWire Protocol

For more information, see Help for the specific instrument.

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Using Protocol Studio : View and Port Setup

View and Port Setup

The View and Port Setup window lets you specify which ports are to be displayed in the Main Grid. You can tailor a display to your own preferences and save the view setup for future use. The window also lets you perform actions on the ports and their modules.

The following topics are available:

- Port/Modules
- View Setups.

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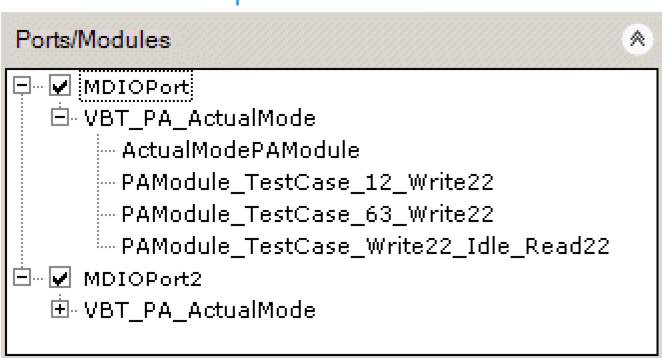
Using Protocol Studio : View and Port Setup : Port/Modules

Port/Modules

The Ports/Modules window lists ports that have been defined for the job and the currently loaded modules for the ports:

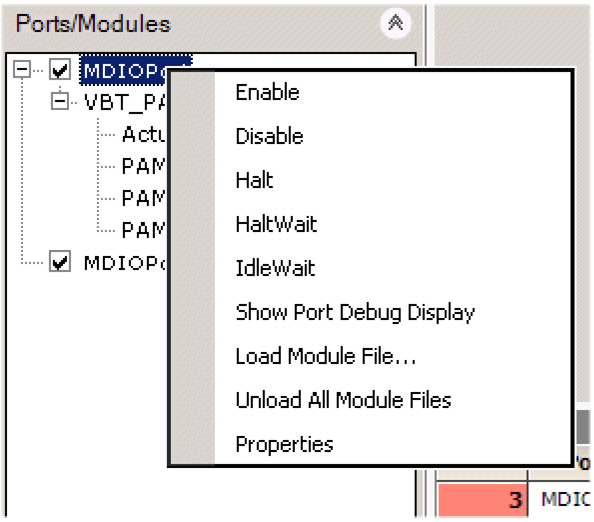


Ports/Modules Window

Ports are the top nodes in the tree structure. A port's module files are the first subnode under the port, and the modules within the file are the subnode under the file.

Port nodes have check boxes by them. Check a box to cause the port to be displayed in the Main Grid. If you uncheck a box, the port is not displayed. Even if a port is not checked, the transaction data is still accumulated and can be displayed in the future by rechecking the box. You can save your selection to a view setup. See View Setups.

Selecting an item in the Ports/Modules window also populates the right-click context menu with actions that you can perform on the item. The following figure shows actions that can be performed on a port:



Ports/Modules Window Context Menu

If you select a module file, you can unload or reload the file. If you select a module in file, you can run the module.

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Using Protocol Studio : View and Port Setup : View Setups

View Setups

Use the View and Port Setup panel to create view setups. A view setup lets you select items that you want included in the display and how you want them displayed, and save the selection as a named setup. You can later view the same display by clicking the name of the setup.

A view setup saves the following information:

- Which ports are checked in the Ports/Modules window of the View and Port Setup panel. This determine the rows that are displayed in the Main Grid.

- Any color, filter, or hide/unhide functions that are applied to the Main Grid. See Column Display Options.

Available setups are listed in the Saved setups window in the View and Port Setup panel. If this is not visible, use View>View and Port Setup to make it visible.

The default built-in setup is Default, which is equivalent to checking all boxes in the Ports/Modules window and not applying any color, filter, or hide/unhide functions. You cannot modify or delete this setup.

To create a view setup, select the items that you want included by checking their boxes in the Ports/Modules window. Set any display options you want. (See Column Display Options.) Then click the Add button. A new string is added to the Saved setups window with the default name NewViewN. The name is in edit mode, so you can change it immediately.

Once you add a setup, you can double-click the name to display the view.

To delete a setup, select it in the Saved setups window and click the Delete button. To delete all user-created setups, right-click in the white area of the Saved setups window (not on a setup name) and select Delete All from the context menu.

 To modify a setup, select it in the Saved setups window. Make changes to the check boxes or display options. Then click the Update button.

To rename a setup, select it in the Saved setups window, right-click, and select Rename. Note that Apply, Delete, and Update are also on the right-click context menu.

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Using Protocol Studio : Saving Transactions to a File

Saving Transactions to a File

Protocol Studio lets you debug transaction specifics offline. You can save captured transactions to a file and then open the file later to continue debugging transactions without the tester.

Protocol Studio translates transactions to XML format and saves them to a file with the .psc extension.

To save the currently captured transactions, use File>Save or Save As. The first time you save to a file, the browser comes up to let you specify a filename and location.

To load a saved file, use File>Open. A browser lets you browse to the saved .psc file.

When you open a file, you are disconnected from IG-XL. As a result, any Protocol Studio function that interacts with IG-XL is disabled:

- Toolbar buttons and menu commands for refreshing and clearing the display are disabled. Buttons and commands for controlling aspects of the display are still enabled.
- Items listed in the Ports/Modules panel do not have right-click context menu options that let you perform actions on them. You can still check and uncheck ports and modules to specify which are to be displayed.
- If there is only one site, the Site control does not appear. If the file contains transactions from multiple sites, the Site control lets you select sites, but the Master button disappears, because the tool cannot determine the master site as set in DataTool.

You can reconnect to IG-XL using the command Tools>Connect to IG-XL. Connecting to IG-XL clears the data displayed from the file.

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